Arista Access Node WFD-ACC-71: Hardware/Environmental Fault Resolution SOP

Node: WFD-ACC-71 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_7269

Description

This Standard Operating Procedure outlines the diagnostic and resolution steps for an Access Node WFD-ACC-71 experiencing a Hardware/Environmental Fault, specifically ASIC memory parity errors, line card crashes, or power supply redundancy loss. The Arista Access Node (Model Number: 7250X-64) employs a modular architecture with redundant power supplies and line cards to ensure high availability and reliability. However, in the event of a hardware fault, the Node's ability to forward traffic is compromised, resulting in the ALARM_7269 alarm being triggered. This SOP is designed to guide technicians through the diagnostic and resolution process, minimizing downtime and ensuring prompt recovery of the affected Node.

Pre-requisite Checks

- Verify that the Whitefield POP has been dispatched with remote hands to assist with the Node's physical inspection.

- Check the redundant link status to ensure that the Node is not under a Link Aggregation Group (LAG) configuration.

- Review the Node's environmental parameters (temperature, humidity, airflow) to identify any potential causes for the hardware fault.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource, Expected Output: System Resource Information:, Description: This command displays the Node's system resource information, including the ASIC memory utilization, which can help identify the root cause of the parity errors.

Command: show interface details, Expected Output: Interface Information:, Description: This command displays detailed information about the Node's interfaces, including the line card status and any errors or alarms associated with the interfaces.

Command: show power supply, Expected Output: Power Supply Information:, Description: This command displays information about the
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Node's power supplies, including the status of the redundant power
supply, which can help identify the root cause of the power supply
redundancy loss.
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Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace asic slot , Description: This command replaces the faulty ASIC in the specified slot. The new ASIC will be automatically configured and online.

Step 2: Step: Replace the faulty line card, Command: replace line-card , Description: This command replaces the faulty line card in the specified slot. The new line card will be automatically configured and online.

Step 3: Step: Replace the redundant power supply, Command: replace power-supply , Description: This command replaces the redundant power supply in the specified slot. The new power supply will be automatically configured and online.

Post-Check Verification

```
Command: show system resource, Expected Output: System Resource
Information:, Description: This command verifies that the Node's
system resource information, including the ASIC memory utilization,
has been restored to normal.

Command: show interface details, Expected Output: Interface
Information:, Description: This command verifies that the Node's
interfaces, including the line card status, have been restored to
normal.

Command: show power supply, Expected Output: Power Supply
Information:, Description: This command verifies that the Node's power
supplies, including the status of the redundant power supply, have
been restored to normal.
```

Roll-back Procedures

- Step: Restore the original ASIC, Command: no replace asic slot , Description: This command restores the original ASIC in the specified slot.

- Step: Restore the original line card, Command: no replace line-card , Description: This command restores the original line card in the specified slot.

- Step: Restore the original power supply, Command: no replace power-supply , Description: This command restores the original power supply in the specified slot.